Significance of GCR1a Binding Site for ADE2 GFP Expression

Purpose

An experiment was done to determine the effects of various mutations in a promoter region on GFP expression and the importance of each transcription factor binding site. The CRISPR Cas9 system was used to knock out the ADE2 gene of yeast and introduce mutations to the promoter region through homology-directed repair (HDR). These mutations were inserted through HDR templates with a single base pair mutation in the GCR1a binding site, the deletion of one transcription factor (TF) binding site alone or in combination with the GCR1a binding site. The experiment collected a pooled data set of resulting GFP expressions to answer a research question on how variable GFP expression is as a result of the single base pair mutation. A research question on the effects each individual TF binding site deletion or in combination with the GCR1a binding site had on GFP expression was also answered, which then helped determine how important each of the TF binding sites were.

Summary

The plasmid pML104, as described by McDonnell *et al.* (2), was chosen for its selectable markers URA3 and AmpR, allowing for cell survival of yeast and E. coli, respectively, on non-uracil and ampicillin plates only when the plasmid is taken up. Colony growth on ampicillin plates from E. coli transformation by Protocol 10 (2) showed that the competent cells took up pML104, allowing for the plasmid's extraction using Protocol 4 (2). A double-digestion using restriction enzymes SwaI and BcII was then done using Protocol 7 (2) to remove the plasmid's scaffold region and its linear conformation was checked by agarose gel electrophoresis. Through Protocol 9b (2), the digested pML104 and a modified gRNA were ligated together to reform the plasmid with gRNA that would guide the Cas9 protein to the cut site on the ADE2 gene. Successful gRNA insertion was determined by perfect sequence alignment of the sample to the expected pML104-gRNA, allowing for yeast transformations with the modified gRNA. The Cas9 protein would have been synthesized by the Cas9 gene in pML104 to create the double-stranded breaks on the gene.

10 HDR templates with varying mutations in the promoter region were used to change the gene sequence through HDR, to knock out the ADE2 gene and to result in some GFP expression. The multiple sequence alignment of Figure 2 visualizes the differences in mutations between templates to help answer the research questions. Yeast transformations were done using Protocol 15 Part 2 (2) where because of the URA3 marker, colonies that grew on the non-uracil plate only had yeast cells that took up the pML104-gRNA and one of the HDR templates. The positive control was yeast plated with pML104 alone to show the expected colony growth. The negative control was yeast plated alone to imply contamination to the samples. Transformation successes were determined by sequence alignments between transformation colonies with good quality control scores and their corresponding HDR template. These implied whether inserted mutations were a result of HDR with perfect alignment or NHEJ with significant mismatches.

Data on the resulting variation of GFP expression was obtained by running one colony of each transformation through flow cytometry. The data was then compared to the baselines of NHEJ and pML104 alone to further determine HDR success.

Conclusion

GFP expression from the HDR5b template, with a single base pair mutation in the GCR1a binding site, was moderately variable in comparison to individual and combined TF binding site deletions. Figure 1 shows that the average GFP expression of HDR5b was 6,847 which is the approximate middle average among the HDR templates given templates with highest to lowest GFP expression: HDR5, HDR5e, HDR5h, HDR5g, HDR5b, HDR5i, HDR5j, HDR5L, HDR5k, and HDR5f. This shows that HDR5b can express some GFP due to assumed successful HDR, but the expression isn't as variable compared to templates like HDR5e and HDR5h. HDR5b's average expression is close to that of HDR5g, with the GCR1a deletion, and the lack of variation despite the difference in mutation could be explained the result of the mutation. HDR5b and HDR5g likely have similar average GFP expressions because the single base pair mutation likely inactivated the entire GCR1a binding site, resulting in expression almost as if the entire GCR1a was deleted.

The deletion of TF binding sites alone causes variable average GFP expression based on which site is deleted, whereas site deletion in combination with GCR1a is more consistent with significant decreased expression. Figure 1 shows that the average GFP expression of HDR5h with the RAP1 binding site deleted was relatively lower than HDR5, the wild type, by 5,114.17. HDR5e with the GCR1b deletion was also relatively lower than HDR5 by 1,466.65. In contrast, HDR5g with the GCR1a site deleted and HDR5i with the TATA box deleted both had significant decreases in expression with differences of 15,423.26 and 18,616.02, respectively, compared to HDR5. This shows that the GCR1a binding site and the TATA box are necessary for high GFP expression, especially since Figure 2 shows that the sites are on the outer ends of the promoter region. The location would affect expression because GCR1a would be recognized first since it comes before GCR1b despite having the same sequence, and the TATA box is before the transcription start site. Figure 1 also shows that the TF binding sites deleted in combination with GCR1a decreases GFP expression consistently with the averages of HDR5j, HDR5k, and HDR5L all below 5k and within the range of GFP expression from HDR5f, with the deleted promoter region. This shows that GCR1a specifically is the most necessary TF binding site, likely because of its specificity, then TATA box likely due to its use of adenine and thymine. RAP1 is of next importance, likely due to its length of 12 nucleotides, and the GCR1b binding site is of least importance likely due to its location and having the same sequence as GCR1a.

Critiques

The resulting data of the pooled GFP showed a few outliers where expression was within range of GFP given off by NHEJ despite the provided HDR template and yeast cells being prone

to HDR. These outliers can be prevented if the experiment was repeated by adding inhibitors of NHEJ in the yeast to prevent cells from using the mechanism. The outliers would have been a result of Protocol 15 Part 2 (2) where the yeast cells use NHEJ instead of HDR regardless of a present template during transformation. This could be changed based on the work of Liu *et al.* (1) where small interfering RNA can be introduced, in that siRNA would be added to the solution plated, that prevents NHEJ from using necessary proteins to continue repair.

The pooled GFP data also showed a few outliers where the GFP expression was in the 100k. This could be explained by Cas9 theoretically making multiple cuts in the gene resulting in multiple inserted HDR templates. The work of Ran *et al.* (3) suggests that using two Cas9 nickases instead of the Cas9 protein would be more efficient in reducing unintended multiple cuts. In Protocol 7 (2), a double-digest was done on pML104 to remove the scaffold but if the Cas9 nickase system is used, then based on the work of Satomura *et al.* (4), the scaffold and the Cas9 gene has to be removed to insert Cas9 nickase and gRNA gene cassettes. These cassettes would eventually express the Cas9 nickase system and create double-stranded breaks reliant on both nicks being made within the same time frame for NHEJ to not occur on one strand.

Figures

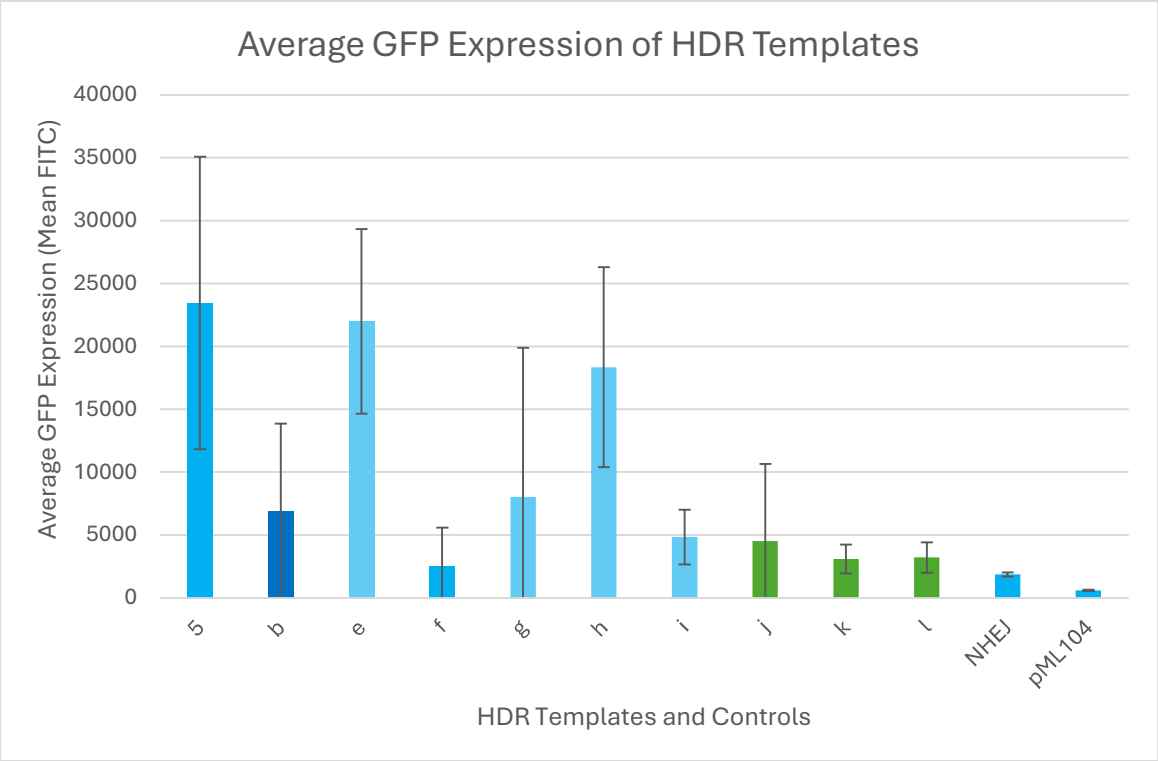

Figure 1: A bar graph showing average GFP expressions of each HDR template. Averages for NHEJ and pML104 act as the control all HDR templates are compared to for determining the success of HDR. Data for templates HDR5 and HDR5f act as controls among the HDR templates to determine variation in GFP expression when compared to the wild type and with no promoter region present, respectively. Error bars show the standard deviation of each individual average.

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HDR Templates                          RAP1 binding site                    GCR1 binding site a
 HDR5>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctggcatccactaaatataatggagcccgctttttaagctg>85
HDR5b>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctggcattcactaaatataatggagcccgctttttaagctg>85
HDR5e>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctggcatccactaaatataatggagcccgctttttaagctg>85
HDR5f>~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~>1
HDR5g>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctg------actaaatataatggagcccgctttttaagctg>79
HDR5h>acagaatatataacatcgta------------aacagtttattcctggcatccactaaatataatggagcccgctttttaagctg>73
HDR5i>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctggcatccactaaatataatggagcccgctttttaagctg>85
HDR5j>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctg------actaaatataatggagcccgctttttaagctg>79
HDR5k>acagaatatataacatcgta------------aacagtttattcctg------actaaatataatggagcccgctttttaagctg>67
HDR5L>acagaatatataacatcgtaggtgtctgggtgaacagtttattcctg------actaaatataatggagcccgctttttaagctg>79

     GCR1 binding site b *           *           *           *           *           *           *
 HDR5>gcatccagaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>170
HDR5b>gcatccagaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>170
HDR5e>------agaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>164
HDR5f>~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~>1
HDR5g>gcatccagaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>164
HDR5h>gcatccagaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>158
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HDR5j>------agaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>158
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HDR5L>gcatccagaaaaaaaaagaatcccagcaccaaaatattgttttcttcaccaaccatcagttcataggtccattctcttagcgcaa>164
                *           *           *           *           *           *           *
 HDR5>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>255
HDR5b>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>255
HDR5e>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>249
HDR5f>~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~>1
HDR5g>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>249
HDR5h>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>243
HDR5i>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>255
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HDR5k>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>237
HDR5L>ctacagagaacaggggcacaaacaggcaaaaaacgggcacaacctcaatggagtgatgcaacctgcctggagtaaatgatgacac>249
                *           *           *           *           *           *           *
 HDR5>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>340
HDR5b>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>340
HDR5e>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>334
HDR5f>~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~>1
HDR5g>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>334
HDR5h>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>328
HDR5i>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>340
HDR5j>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>328
HDR5k>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>322
HDR5L>aaggcaattgacccacgcatgtatctatctcattttcttacaccttctattaccttctgctctctctgatttggaaaaagctgaa>334
                *           *           *           *                   TATA Box    *           *
 HDR5>aaaaaaggttgaaaccagttccctgaaattattcccctacttgactaataagtatataaagacggtaggtattgattgta>420
HDR5b>aaaaaaggttgaaaccagttccctgaaattattcccctacttgactaataagtatataaagacggtaggtattgattgta>420
HDR5e>aaaaaaggttgaaaccagttccctgaaattattcccctacttgactaataagtatataaagacggtaggtattgattgta>414
HDR5f>~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~>1
HDR5g>aaaaaaggttgaaaccagttccctgaaattattcccctacttgactaataagtatataaagacggtaggtattgattgta>414
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HDR5L>aaaaaaggttgaaaccagttccctgaaattattcccctacttgactaataag--------gacggtaggtattgattgta>406
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Figure 2: A multiple sequence alignment is shown of all 10 HDR templates used that were trimmed to start from 20 base pairs upstream of the RAP1 binding site to 20 base pairs downstream of the TATA box. The HDR templates differ in deletions of one transcription factor binding site or a deletion of one transcription factor binding site in combination with the GCR1a site. HDR5 differs in that it has the original promoter sequence, acting as the wild type, and HDR5f differs in that the entire promoter sequence was deleted. HDR5b differs in having a single base pair mutation in the GCR1a binding site rather than a complete site deletion. Regions of the sequence alignments highlighted in red are mismatches of the HDR template to HDR5 whether due to deletions or nucleotide mutations.

References

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